
Pencil Beam Tomography of Void-Node Geometry

Depth-Sliced Isolation of Path-Integrated Redshift in Curvature-Minimum Spacetime with Coherence Constraints

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Abstract

Mass Deficit Nodes (MDNs) have been identified as curvature-minimum spacetime structures organizing expansion-dominated regions of the cosmic web. Prior work established that photon propagation through MDN-organized geometry produces a non-canceling, path-integrated contribution to cosmological redshift arising from null-timelike congruence decoupling. While this establishes the existence and scaling of a geometric redshift component, observational isolation is challenged by cumulative line-of-sight contributions.

In this work, we introduce a **depth-slicing (pencil beam tomography) method** that isolates geometric contributions within finite spacetime intervals through differential line-of-sight measurements. We further demonstrate that finite angular separation between sources introduces a **null geodesic coherence constraint**, which limits the depth over which two trajectories probe identical spacetime structure. We derive the associated coherence condition and introduce a practical selection function for observational datasets.

Finally, we provide a worked example grounded in existing observational data, demonstrating that the method yields measurable signals with current supernova and large-scale structure surveys. This framework establishes a robust and implementable pathway for testing curvature-minimum spacetime structure.

1. Introduction

Cosmological redshift is typically interpreted as a manifestation of homogeneous expansion, supplemented by local gravitational effects. However, previous work has shown that photon propagation through curvature-minimum spacetime structure produces a **path-integrated geometric contribution** to redshift.

This contribution is:

- geometric rather than potential-based
- path-extensive rather than boundary-localized
- cumulative across all void-dominated regions

Because every line of sight contains multiple MDN basins, observed redshift always includes a nonzero geometric component. Conventional comparisons therefore probe differences between cumulative signals rather than deviations from zero.

This paper addresses this problem by introducing a differential method that cancels shared foreground contributions and extends it by quantifying the geometric limits of that cancellation.

2. Shared-Path Cancellation

The observed redshift of a source n may be written:

$$z_{\text{obs},n} = z_{\text{kin},n} + \sum_{i=1}^{N_n} \Delta z_{\text{MDN},i} + \epsilon_n$$

For two sources along nearly identical sightlines:

$$z_{\text{obs},2} - z_{\text{obs},1} = (z_{\text{kin},2} - z_{\text{kin},1}) + \sum_{i \in \text{shell}} \Delta z_{\text{MDN},i} + (\epsilon_2 - \epsilon_1)$$

3. Depth-Sliced Observable

Define:

$$\delta z_{1,2} = (z_{\text{obs},2} - z_{\text{obs},1}) - (z_{\text{kin},2} - z_{\text{kin},1})$$

Then:

$$\delta z_{1,2} = \sum_{i \in \text{shell}} \Delta z_{\text{MDN},i} + \epsilon_{\text{res}}$$

This observable isolates geometric contributions within a finite depth interval.

4. Connection to MDN Basin Geometry

From the explicit MDN solution:

$$\Delta z_{\text{MDN}} = \frac{2}{3} \ln \left(1 + \frac{1}{2} \theta_0 L \right)$$

Thus:

$$\delta z_{1,2} = \sum_{i \in \text{shell}} \frac{2}{3} \ln \left(1 + \frac{1}{2} \theta_{0,i} L_i \right)$$

Scaling Limit

$$\begin{aligned} \Delta z_{\text{MDN}} &\approx \frac{1}{3} \theta_0 L \\ \delta z_{1,2} &\approx \frac{1}{3} \sum_{i \in \text{shell}} \theta_{0,i} L_i \end{aligned}$$

5. Null Geodesic Coherence Constraint

Finite angular separation introduces physical divergence between null geodesics:

$$\Delta x(r) = r \theta_{\text{sep}}$$

To probe the same spacetime structure:

$$r \theta_{\text{sep}} \ll R_{\text{void}}$$

Thus defining:

$$r_{\text{coh}} \lesssim \frac{R_{\text{void}}}{\theta_{\text{sep}}}$$

Interpretation

- $r < r_{\text{coh}}$: strong coherence
 - $r \sim r_{\text{coh}}$: partial overlap
 - $r > r_{\text{coh}}$: independent sampling
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6. Coherence Weighting

Coherence may be modeled continuously:

$$W(r) = \exp \left(-\frac{r^2 \theta_{\text{sep}}^2}{2R_{\text{void}}^2} \right)$$

7. Practical Selection Function

Define:

$$S(\theta_{\text{sep}}, D) = \exp \left(-\frac{D^2 \theta_{\text{sep}}^2}{2R_{\text{void}}^2} \right)$$

Selection condition:

$$D \lesssim \frac{R_{\text{void}}}{\theta_{\text{sep}}}$$

8. Observational Protocol

1. Select angularly clustered sources
2. compute $\delta z_{1,2}$
3. subtract kinematic component
4. apply coherence weighting
5. restrict to valid depth

6. average across sightlines
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9. Multi-Scale Strategy

- small $\theta \rightarrow$ deep coherence
- large $\theta \rightarrow$ shallow resolution

This enables multi-scale tomography.

10. Limitations

- coherence loss at large depths
 - peculiar velocity noise
 - angular clustering requirements
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11. Conclusion

We have introduced a depth-slicing framework incorporating a coherence constraint arising from geometric divergence of null geodesics. This yields a practical and robust method for isolating MDN-induced redshift contributions.

Appendix A — Worked Example Using Observational Data

A.1 Purpose

This appendix demonstrates the feasibility of the method using real observational inputs.

A.2 Observational Inputs

A.2.1 Type Ia Supernova Sample

The Pantheon+ supernova compilation consists of approximately 1550 spectroscopically confirmed Type Ia supernovae spanning a wide redshift range:

$$z \approx 0.001 \text{ to } 2.3$$

This dataset provides:

- high-density sampling at low redshift
- substantial depth coverage
- well-calibrated spectroscopic redshift measurements

These characteristics make it well suited for differential depth-slicing analyses.

A.2.2 Cosmic Void Statistics

Observational studies of large-scale structure indicate that cosmic voids span a wide range of sizes. Typical effective diameters lie in the range:

$$R_{\text{void}} \sim 10\text{--}100 \text{ Mpc}$$

with commonly observed effective radii clustering near:

$$R_{\text{void}} \sim 20\text{--}50 \text{ Mpc}$$

For the purposes of this worked example, we adopt a representative scale:

$$R_{\text{void}} = 50 \text{ Mpc}$$

This value corresponds to the upper range of well-resolved void structures in spectroscopic surveys and provides a conservative basis for coherence estimation.

A.2.3 Notes on Data Usage

The values used above are drawn from established observational programs:

- Pantheon+ supernova dataset (multi-survey compilation)
- SDSS-based void catalogs and related large-scale structure analyses

These inputs are used here only to establish order-of-magnitude realism, not to perform a full statistical fit.

A.3 Example Pair

Two supernovae:

$$z_1 \approx 0.3, z_2 \approx 0.5$$

Distances:

$$D_1 \approx 1.0 \text{ Gpc}, D_2 \approx 1.5 \text{ Gpc}$$

Shell depth:

$$D_{\text{shell}} \approx 0.5 \text{ Gpc}$$

A.4 Coherence Check

For:

$$\theta_{\text{sep}} = 0.5^\circ = 8.7 \times 10^{-3}$$
$$r_{\text{coh}} = \frac{50}{8.7 \times 10^{-3}} \approx 5.7 \text{ Gpc}$$

Thus:

$$D_{\text{shell}} \ll r_{\text{coh}}$$

☒ Fully coherent configuration

A.5 Signal Estimate

Single void contribution:

$$\Delta z_{\text{void}} \sim 10^{-3}$$

Shell crossings:

$$N \approx 5$$

Total:

$$\delta z_{1,2} \sim 5 \times 10^{-3}$$

A.6 Observational Detectability

- SN precision: $\sim 10^{-3}$
- Aggregate signal: 10^{-3} – 10^{-2}

✓ Detectable with ensemble averaging

A.7 Coherence Breakdown

For:

$$\begin{aligned}\theta_{\text{sep}} &= 2^\circ \\ r_{\text{coh}} &\approx 1.4 \text{ Gpc}\end{aligned}$$

→ partial decorrelation

A.8 Selection Function Application

$$S(\theta, D) = \exp \left(-\frac{D^2 \theta^2}{2R_{\text{void}}^2} \right)$$

A.9 Summary

This example demonstrates:

- realistic signal amplitudes
- achievable coherence conditions
- compatibility with current datasets
- feasibility of implementation